

MICROSERVICES COURSE · SPRING BOOT & SPRING CLOUD

Professional Edition

Program Kickoff

Welcome · Roadmap · Platform · Working Model · Assessment



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ITSharks

IT Learning & Training

Why Microservices Matter Right Now



Netflix

Deploys to production
100s of times per day



Amazon

Deploys a new version
every 11.7 seconds



Uber

Runs 2,000+ microservices
in production

Industry leaders such as Netflix, Amazon, and Uber adopted microservices to achieve independent deployment, scalability, and team autonomy.

The ability to deploy independently, scale selectively, and own bounded domains is now a baseline expectation in any serious engineering organisation.

This course trains you to design, build, and operate systems the way industry leaders do.

Problems This Course Equips You to Solve

 Problem

A deploy in one module
breaks the entire system

 Solution

Independent deployable services

 Problem

One slow service
exhausts all threads

 Solution

Resilience4j: CB + Bulkhead + TimeLimiter

 Problem

Payment fails midway —
order is stuck in limbo

 Solution

Saga Pattern + Compensation

 Problem

8 services use
8 different configs

 Solution

Spring Cloud Config Server

 Problem

Order can't find
Inventory's address

 Solution

Eureka Service Discovery

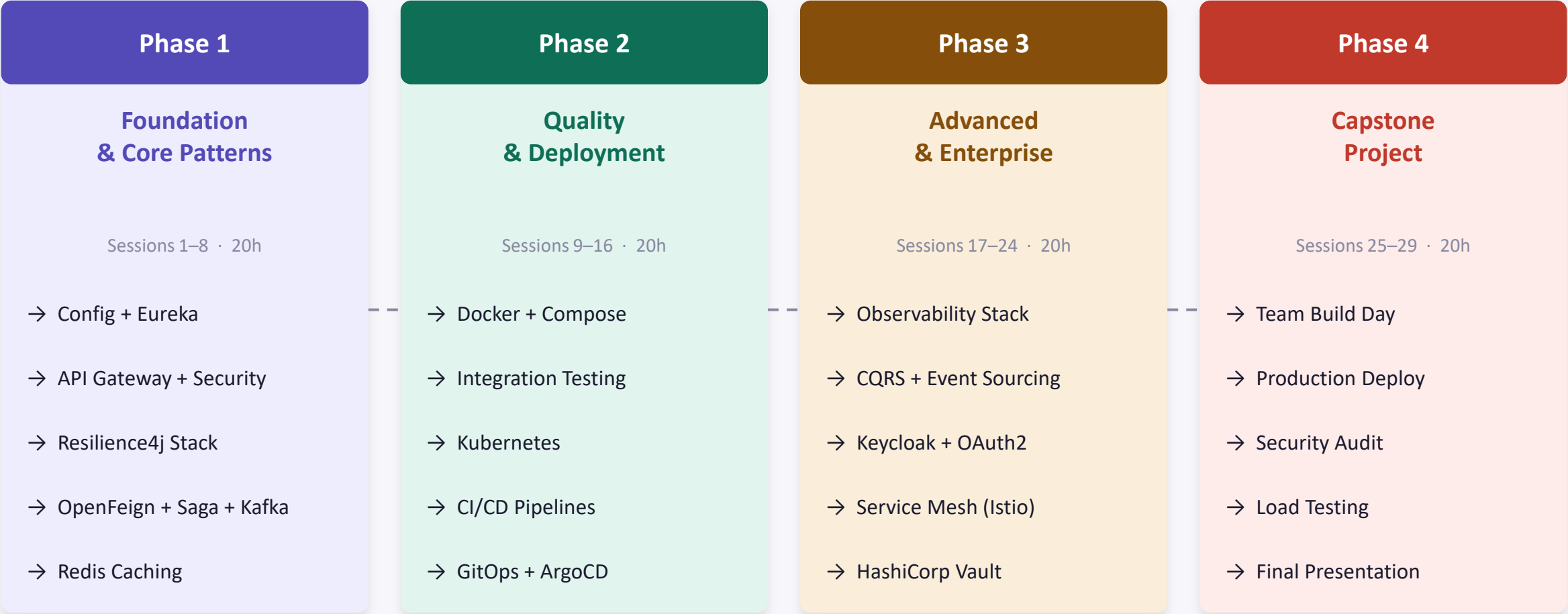
 Problem

1000 requests/min
collapse the database

 Solution

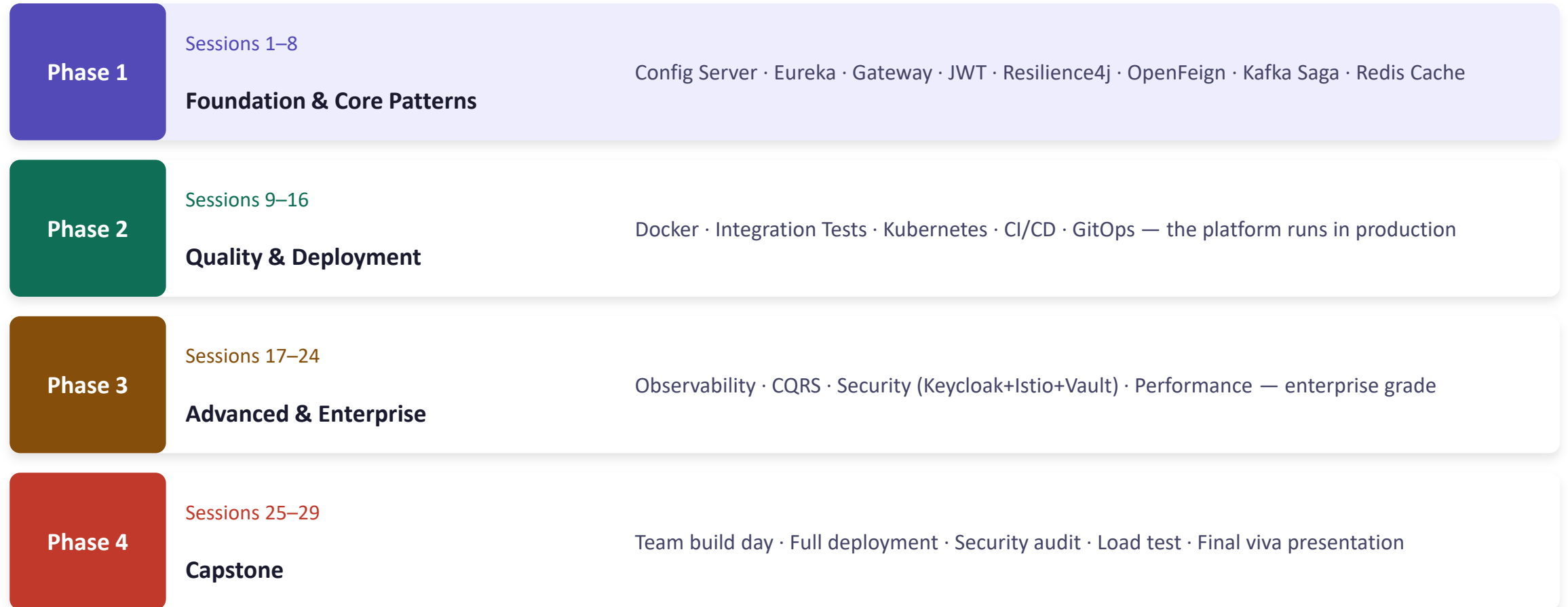
Redis Caching + Cache-Aside

29 Sessions — 80 Hours — 4 Phases



4 Offline Anchor Days (5h each) — Sessions 1, 9, 17, 25 · 25 Online Sessions (2.5h each) — Mon/Wed 3:00-5:30 PM

From Foundation to Production — The Journey



You Build ONE Real Platform — Session 1 to Session 29

Enterprise E-Commerce Platform

A production-grade microservices system you will design, build, secure, test, and deploy.
Not a toy example. Not isolated exercises. One growing system — every session adds a real capability.



Every session has purpose

Your lab isn't a practice file — it's part of a real product you're building



Architecture decisions persist

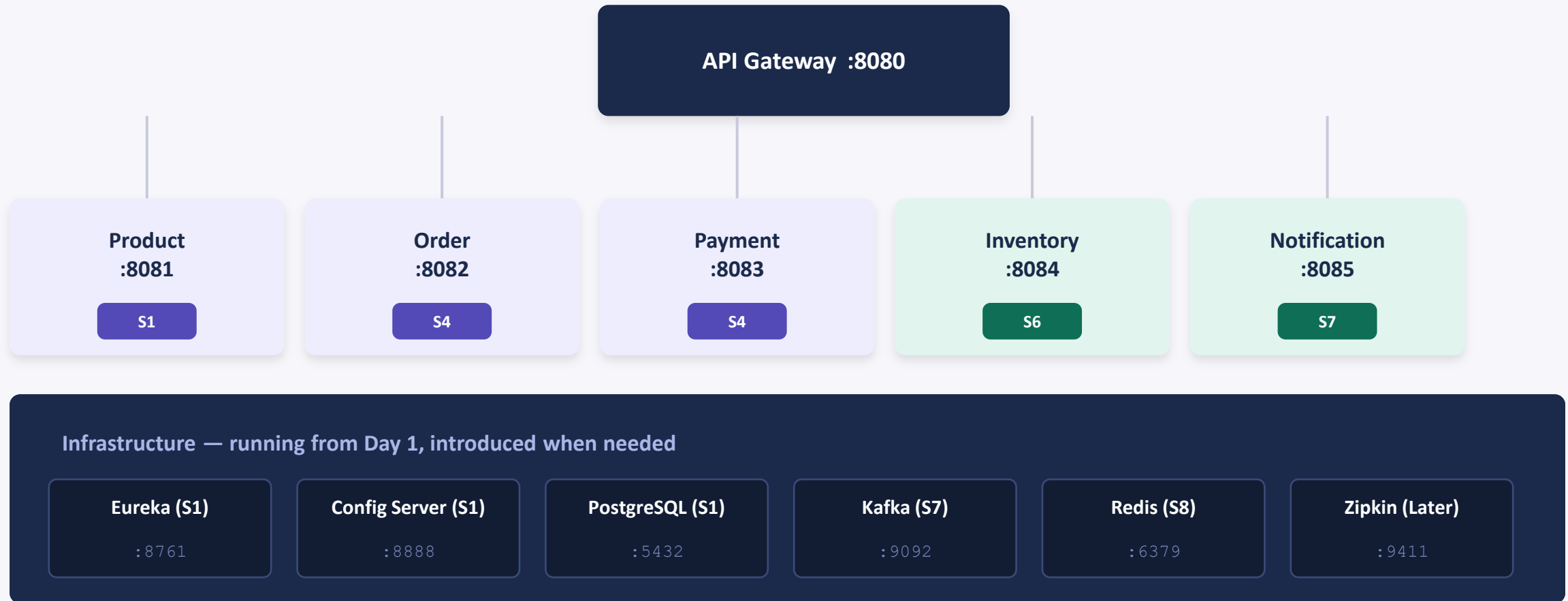
A bad design in Session 3 costs you in Session 7 — just like real engineering



Ready for production

By Session 29, what you built can actually run in a real cloud environment

Platform Architecture — Full Picture



microservices-pro-platform · Branch: main (your work) + reference (instructor implementation)

The Continuous Capstone — Why It Matters

Traditional Training

New exercise every session — discarded after

Lab 3 has no relation to Lab 7

"I built a circuit breaker demo"

Capstone starts from scratch in the final sessions

Low motivation — throwaway work

Continuous Capstone (This Course)

One evolving platform — every lab builds on the last

A design choice in Session 3 affects Session 7 — by design

"I added resilience to our Order Service"

Capstone = completion of what you already built

High motivation — real product, real decisions

How the Course Is Delivered

Offline Anchor Day

4 Sessions (1, 9, 17, 25)

Wednesday · 10:00 AM – 3:00 PM

5 Hours

- Deep-dive theory + whiteboard
- Extended live coding (no rush)
- Full supervised lab (90 min)
- Architecture discussion + quiz

Online Live Session

25 Sessions

Monday & Wednesday · 3:00 – 5:30 PM

2.5 Hours

- Knowledge Check on pre-reading
- Concept + live coding (60 min)
- Lab work — independent (35 min)
- Quiz + Checkpoint Commit

How You Will Learn — The Session Loop

1 Pre-Reading

15 min before each session.
Targeted material — not optional.

2 Knowledge Check

3 questions on the pre-reading,
called by name at session start.

3 Concept + Problem

The problem is introduced first.
Always. Then the solution.

4 Live Coding

Trainer codes live — you follow
and type along. Real code, real
mistakes.

5 Lab

You implement independently. 35
minutes of focused building.

6 Quiz

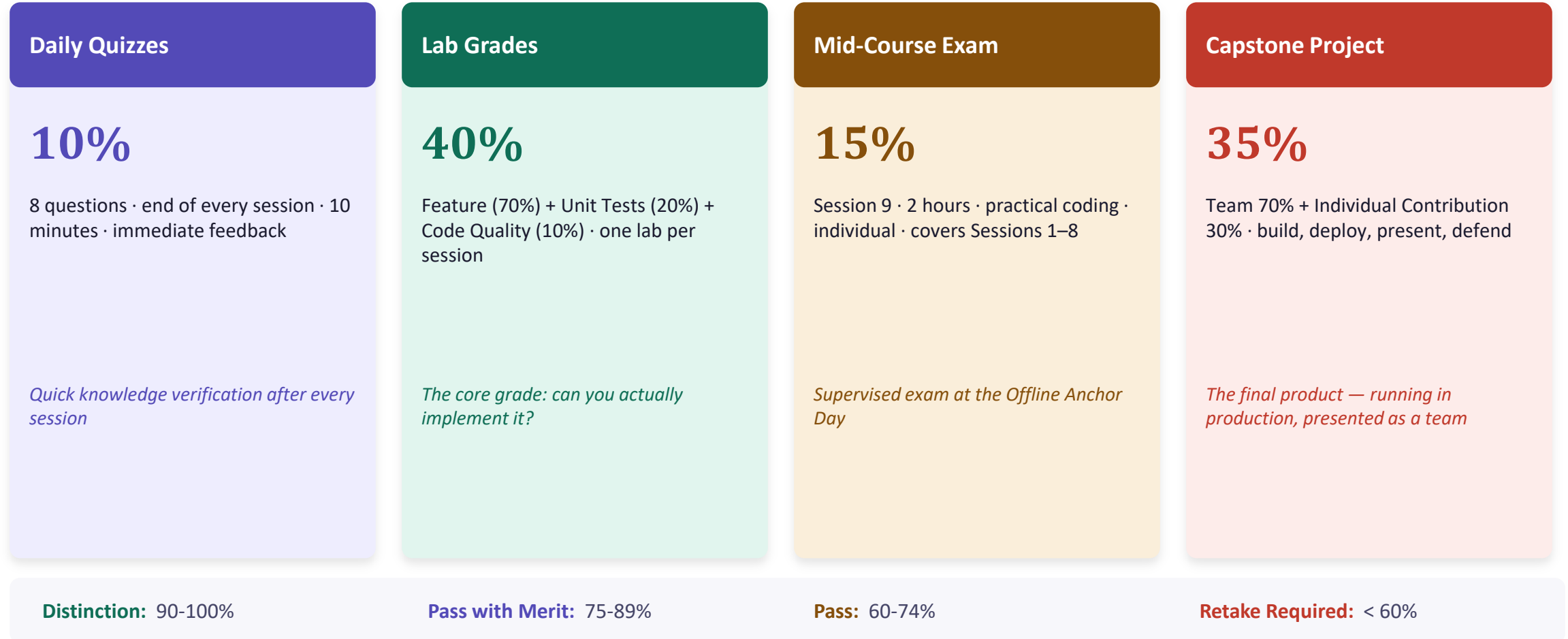
8 questions. 10 minutes.
Immediate feedback on the
session.

7 Checkpoint Commit

Commit and push your work. No
commit = no grade for that lab.

🔄 *This loop repeats for all 29 sessions*

Assessment Model — Four Components



Expectations & Commitments

What We Expect From You

- **Pre-reading every session**
15 min is enough — but it must be done
- **Labs completed between sessions**
Don't wait for the next session to finish
- **Checkpoint commit pushed**
No commit = no grade, no exceptions
- **Unit tests with every lab**
Tests are not optional — 20% of your grade
- **Pre-reading before Knowledge Check**
You will be called on by name, every session

What You Get From Us

- **Full Session Packs**
Detailed instructor guide + code + labs for every session
- **Reference branch**
Instructor implementation you can compare against
- **Architecture Clinics**
Two full reviews (Sessions 8 & 24) of your design decisions
- **Assessment Bank**
64 exam questions with explanations — all mapped to traps
- **Production-grade patterns**
Patterns that work in real systems, not just demos



Kickoff Complete

Let's Start Session 1

Architecture & Spring Cloud · Service Discovery · Centralized Configuration

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